Chair of Computer Vision and Artificial Intelligence
School of CIT
Technical University of Munich

E0092
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Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
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Computer Vision III: Detektion, Segmentierung und Tracking

Exam:	IN2375 / Endterm	**Date:**	Friday 27th February, 2026
Examiner:	Prof. Dr. Daniel Cremers	**Time:**	08:00 – 09:30

Working instructions

- This exam consists of **16 pages** with a total of **5 problems**.
  Please make sure that you received a complete copy of the exam.
- In this exam you can receive a total of **60 credits**.
- Detaching pages from the exam is strictly forbidden.
- Allowed resources:
  - one **analog** dictionary English ↔ native language
- Other resources (e.g. cheatsheets) are forbidden. Cheating in the exam results in exam failure.
- Physically turn off all electronic devices and put them out of your sight (e.g. into your bag).
- Write clearly and be concise. If you believe a question is ambiguous, state your interpretation and answer the question under this interpretation.
- Do not use pencils. Do not write in red or green ink.
- Circled numbers ⓧ denote the credits you can earn per question.
- Good luck!

Left room from _____ to _____ / Early submission at _____

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Problem 1 Single Choice Questions (12 credits)

a) Select **one** option that applies to the Viola-Jones detector:

- ☐ It is sensitive to different identities
- ☐ It is an objectness-based approach
- ☐ It is sensitive to face orientation
- ☐ It is a template-based approach

b) Select **one** option that applies to the Overfeat detector:

- ☐ Classification and Localization run only on locations with high objectness
- ☐ It does not need Non-Maximum Suppression
- ☐ Classification and Localization are trained in two different steps
- ☐ Classification and Localization need two different feature extractors

c) Select **one** option that applies to Selective Search:

- ☐ It is a learning-based technique to detect objectness in an image
- ☐ It is used as post-processing in object detection pipelines
- ☐ Reduces the workload for object detectors
- ☐ It starts by considering the whole image a single region, and progressively breaking it down into smaller pieces

d) Select **one** option that applies to YOLOv1:

- ☐ It includes RoI pooling, making it faster than the R-CNN family
- ☐ The upper bound for the number of detected objects in an image is equal to the number of image pixels
- ☐ It requires the input image to be rescaled and warped to fit a fixed dimension
- ☐ YOLO tends to predict fewer boxes than the R-CNN family, and so it does not need Non-Maximum Suppression

e) Select **one** option that applies to Spatial Transformer Networks (STNs):

- ☐ They use self-attention
- ☐ They can be used in the feature space
- ☐ STNs require ground truth transformation labels to learn
- ☐ Having a backbone that uses more than one STN would not have advantages

f) Select **one** option: If two images p and q belong to a space equipped with a metric d, then for any p and any q it must hold:

- ☐ $d(p, q) = d(q, p)$
- ☐ $d(p + z, q + z) = d(p, q)$
- ☐ $d(p, p) > 0$
- ☐ $d(p, q) > 0$

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g) Select **one** option: Consider the Hinge Loss we have seen during lecture. For $m = 2$, $f(A) = 0.4$, and $f(B) = 0.9$ with (A, B) a negative pair, the value of the loss is:

- ☐ 0.25
- ☐ 3.75
- ☐ 1.75
- ☐ 0

h) Select **one** option: What is true about online and offline tracking?

- ☐ Online tracking makes use of future frames; offline tracking relies on the context of previous frames.
- ☐ Offline tracking is used in real-time applications.
- ☐ Online tracking can also be adapted for offline tracking.
- ☐ Online tracking can handle long-term occlusions effectively.

i) Select **one** option that applies to the Atrous Convolution:

- ☐ the dilation rate hyperparameter changes the number of learnable parameters
- ☐ it is used as an upsampling technique
- ☐ the receptive field of an atrous convolution with kernel 5 and rate 2 is 9
- ☐ by stacking a sequence of many atrous convolutions of the same rate, we obtain multi-resolution features

j) Select **one** option that applies to DINO:

- ☐ at inference time, both the teacher and the student are used
- ☐ its features have geometrical emergent properties, *e.g.*, characterizing left from right
- ☐ gradient is propagated through the teacher but not through the student
- ☐ it is trained unsupervisely

k) Select **one** option that applies to Virtual Adversarial Networks:

- ☐ They corrupt the input with arbitrarily severe noise, and encourage the network output to be proportionally shifted
- ☐ They corrupt the input with arbitrarily severe noise, and encourage the network output to be unchanged
- ☐ They corrupt the input with limited noise, and encourage the network output to be proportionally shifted
- ☐ They corrupt the input with limited noise, and encourage the network output to be unchanged

l) Select **one** option that applies to Segment Anything:

- ☐ It solves instance segmentation
- ☐ In its last training stage, annotators focus on outliers that the model does not detect
- ☐ Points, boxes, and texts are all valid input prompts
- ☐ In its training, no labeled data have been used

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Problem 2 **Detection (12 credits)**

0 1 2

a) One of the traditional approaches to object detection is based on template matching with sliding windows. Outline how it operates (1.0)

Name one disadvantage of template matching with sliding windows. (1.0)

0 1 2 3

b) What is the equation of the Focal Loss (1.0) and what problem of Cross Entropy Loss does it solve? (1.0)

The Focal Loss has one parameter that need to be set. What is its role? What can happen if it is set too high? (1.0)

0 1 2

c) What element of the R-CNN architecture requires resizing the input crops to a fixed size? (1.0)

How does Fast R-CNN overcome this? (1.0)

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d) Describe the AP and the algorithmic steps to compute it. (1.5)

0
1
2
3

How does mAP differ? (0.5)

Sometimes the AP metric comes with a number (*e.g.*, $AP^{0.75}$). What does it mean? (1.0)

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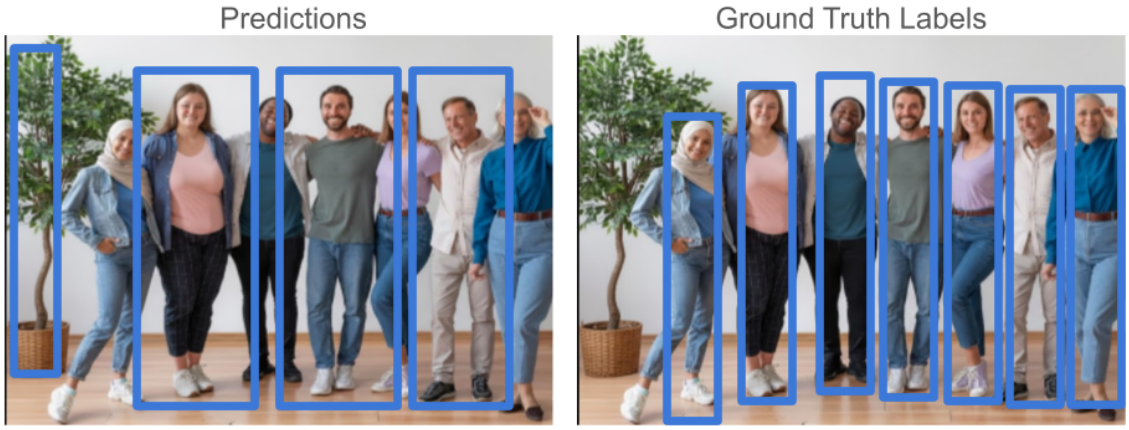
0 1 2

e) The image represents the predictions of a person detector network and the ground truth bounding boxes. Compute the Precision and the Recall. (1.0)



Assume the false negatives are due to the Non-Maximum Suppression step that kills too many correct predictions. What could be the cause, and then the fix? (1.0)

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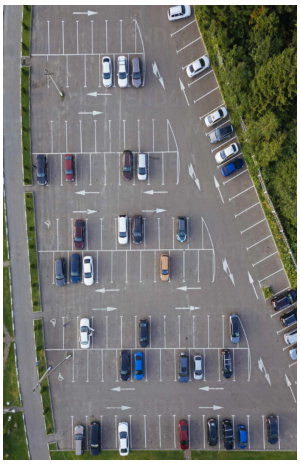
Problem 3 Tracking (12 credits)

a) Your friend Otto owns a parking lot. He wants to build a vision system to track the cars, so he buys a black-and-white camera and installs it, filming from above (see the picture). He believes that a picture of the empty parking lot would help track the cars. However, Even though cars do not stay parked for long (from a few minutes to an hour), over weeks of video footage, there is never a single frame without cars. How can Otto use SVD to isolate the background? Describe the steps. 3.0

0 1 2 3



b) After some research, Otto decided to try deep learning approaches. He collected a dataset of daytime videos, annotated several cars, and trained a network. At test time, he noticed that cars are accurately detected in the first frame they appear, but as daylight changes, the tracks are incorrectly re-identified. How can Otto solve this issue at inference time, without labeling new images? Name the technique and describe the steps. 3.0

0 1 2 3

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0 1 2 3

c) Otto wants to compare the results of the old and new video trackers in terms of overall accuracy. Which metric can he use? Specify the metric name (1.0), write the equation (1.0), and explain its terms (1.0).

0 1 2 3

d) Otto reads that the Tracktor method does not need video data for training. Explain how Tracktor inference works (1.5), which motion model prior it uses (0.5), and why it does not need videos for training (1.0)

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Problem 4 Segmentation (12 credits)

a) In Spectral Clustering for image segmentation, what are the input features of the K-Means step? (1.0)

0 1 2 3

How are those features obtained? Describe the steps. (1.0)

Why not use directly the pixel intensity values as input to K-Means? (1.0)

b) Explain the computation performed by the Correlation Layer used in FlowNet. (1.0) Consider a correlation layer L that takes as input two features of size $H \times W$ with C channels each. What is the dimensionality of the output? (0.5)

0 1 2

How many learnable parameters does L have? (0.5)

c) Give the equation of triplet loss (1.0) and explain its logic (1.0)

0 1 2

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0 1 2

d) What is panoptic segmentation (1.0)?

For which object categories do we want semantically consistent features, and for which ones instance-aware features? (1.0)

0 1 2 3

e) What are the two branches of PanopticFCN architecture, and what do they predict? (1.0)

What is the loss used to train the Position Head? (1.0)

What is the purpose of the kernel fusion step, and how is it performed? (1.0)

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Problem 5 **Recent Research (12 credits)**

a) How many parameters are learnable in a self-attention layer where Query, Key, and Value are processed by linear layers of size 64, and the input tokens have an embedding dimension of 128? (Reporting the terms you would sum up and explaining why is enough; there is no need to provide the exact number) (2.0)

0 1 2 3

How does the number of parameters change if we use multi-heads with four branches? (1.0)

b) What is the role of the [class] token in the ViT? (1.0)

0 1

c) How is GIoU defined? Provide the equation and explain its terms. (2.0) Why do we prefer GIoU over IoU as a loss to train a DETR model? (1.0)

0 1 2 3

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0 1 2 3

d) What is the purpose of Linear Probing in unsupervised learning? (1.0)

Describe how Linear Probing works and what is the intuition behind it (*i.e.*, why is it used). (2.0)

0 1 2

e) In the lectures, we have discussed how learned representations balance between removing useless information (compression) while retaining the information relevant for the target task (expressivity). Write the Information Theory equation we have seen to model such a trade-off, explain its variables and terms (2.0).

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Additional space for solutions–clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

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